

BARRIE, ON, CanadaWMO: **714361**Lat: **44.380N**Lon: **79.780W**Elev: **968**StdP: **14.19**Time Zone: **-5.00 (NAE)**Period: **94-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-2.2	-18.9	2.0	-7.2	-12.1	3.0	-1.3	22.2	20.2	20.1	22.7	5.7	110	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	85.8	71.8	82.5	70.3	79.4	68.0	74.0	82.6	71.9	80.0	69.9	76.9	7.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	119.2	79.2	69.1	111.0	76.0	67.4	104.3	73.9	38.2	83.0	36.2	79.8	34.5	76.8	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.8	15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB													
			WB	-17.9	76.6	7.6	3.9	-23.4	79.5	-27.9	81.8	-32.1	84.0	-37.6	86.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	18.9	21.2	29.0	41.3	52.6	63.3	66.9	65.7	58.7	47.3	37.3	25.2
	DBStd	19.16	12.35	10.83	10.98	9.14	8.19	7.63	5.71	5.50	7.55	7.89	8.42	10.58
	HDD50	4106	965	807	656	290	65	4	0	0	13	151	386	769
	HDD65	7935	1430	1227	1115	714	395	120	45	59	217	550	830	1233
	CDD50	1939	0	0	6	27	145	403	525	486	273	66	7	1
	CDD65	293	0	0	0	2	10	69	106	79	27	0	0	0
	CDH74	2347	0	0	0	23	101	607	808	584	215	9	0	0
	CDH80	560	0	0	0	4	11	178	188	137	42	0	0	0
(14) Wind	WSAvg	6.5	7.6	7.6	7.5	7.7	6.9	5.6	5.0	4.3	5.1	6.0	7.3	7.1
(15) Precipitation	PrecAvg	32.8	2.4	2.0	2.2	2.6	2.9	3.1	3.2	3.0	3.0	2.8	2.9	2.7
	PrecMax	41.8	4.1	4.2	5.7	5.2	5.2	8.1	4.7	6.6	5.3	5.3	5.4	4.3
	PrecMin	24.5	1.1	1.0	0.7	0.9	1.0	0.8	1.3	1.4	1.2	0.9	1.0	1.2
	PrecStd	3.9	0.8	0.9	1.1	1.1	1.0	1.6	1.1	1.1	1.0	1.1	1.1	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	48.0	65.9	77.7	81.4	88.3	89.2	89.8	85.3	75.0	61.6	54.5
		MCWB	45.0	45.1	56.0	61.7	67.3	74.4	75.2	72.1	70.2	64.4	54.3	51.8
	2%	DB	40.7	43.3	56.9	68.6	76.8	85.2	85.0	83.3	79.7	69.0	57.6	47.4
		MCWB	38.8	39.6	50.3	58.9	64.3	71.7	72.1	71.1	67.8	60.8	54.0	45.2
	5%	DB	37.3	39.5	49.3	61.8	72.5	81.5	81.6	79.8	75.7	64.3	54.1	42.3
		MCWB	35.4	36.7	43.0	51.3	61.4	69.6	69.9	68.5	66.1	58.7	50.9	40.5
	10%	DB	34.2	35.4	43.3	56.0	67.4	76.9	78.5	76.9	71.3	60.1	50.2	38.2
		MCWB	32.8	33.1	39.4	46.9	57.9	66.8	67.5	67.5	63.4	55.0	47.5	36.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	44.8	58.9	65.4	69.3	76.1	76.8	75.4	73.1	65.4	57.5	52.0
		MCDB	45.7	46.4	64.4	73.9	77.9	86.3	85.9	83.4	79.7	72.1	60.1	53.6
	2%	WB	39.3	40.0	50.7	59.2	66.0	73.1	74.3	72.8	69.7	61.8	54.7	45.9
		MCDB	40.3	42.8	56.5	66.7	74.9	83.2	82.9	80.3	76.4	67.3	57.1	47.4
	5%	WB	35.7	37.0	44.0	53.1	62.7	70.6	71.6	70.7	67.1	58.9	51.0	40.6
		MCDB	36.9	39.7	48.4	60.1	71.0	79.3	79.4	77.5	73.0	63.7	53.3	42.1
	10%	WB	32.9	33.4	39.1	48.2	58.9	68.3	69.4	68.7	65.0	56.0	47.1	36.3
		MCDB	34.3	35.4	43.4	55.1	65.0	74.8	75.9	74.4	70.2	59.7	50.1	38.2
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	14.8	15.3	17.8	19.5	18.8	19.1	19.2	18.8	16.1	12.2	12.1
		MCDBR	12.1	15.6	22.0	28.3	24.5	24.0	22.8	23.6	24.4	22.9	17.7	14.3
		MCWBR	11.3	13.5	16.2	18.8	15.5	13.3	12.1	12.3	14.3	15.6	15.0	12.1
		MCDBR	11.6	15.1	21.5	25.3	22.5	21.3	20.4	19.5	21.2	21.2	16.4	12.8
	5% WB	MCWBR	11.2	13.8	17.1	18.6	15.7	12.7	12.0	11.7	13.3	15.6	14.9	12.0
(40) Clear-Sky Solar Irradiance	taub		0.280	0.295	0.315	0.351	0.382	0.402	0.399	0.391	0.358	0.340	0.307	0.280
	taud		2.381	2.323	2.410	2.365	2.331	2.292	2.321	2.332	2.423	2.461	2.502	2.443
	Ebn at Noon		268	284	293	289	282	275	275	273	274	265	258	258
	Edh at Noon		24	30	32	36	39	41	39	37	32	27	22	21
(44) All-Sky Solar Radiation	RadAvg		431	709	1123	1423	1711	1867	1888	1624	1281	747	444	313
	RadStd		49	89	118	141	172	138	122	85	122	79	62	38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A